MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ADVANCED SCHOTTKY TTL,

DECODERS, MONOLITHIC SILICON

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, advanced Schottky TTL, decoder microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Decoder, 1 of 8, active low outputs with enable inputs.
02	Dual decoder, 1 of 4, active low outputs with enable inputs.

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8") dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8") flat package
X	C-2 (20-terminal, .350" x .350") square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 to 7.0 V dc
Input voltage range	-1.2 V dc @ -18 mA to 7.0 V dc
Storage temperature range	-65°C to 150°C
Maximum power dissipation (P_D) 1/	
Device type 01	110 mW
Device type 02	110 mW
Lead temperature (soldering, 10 seconds)	300°C

1/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Thermal resistance, junction to case θ_{JC}:

Case E	50°C/W
Case F	70°C/W
Case X	60°C/W
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions.

Supply voltage	4.5 to 5.5 V dc
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.8 V dc
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Case outlines. The case outlines shall be as specified in 1.2.3.

3.2.2 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figures 1 and 2.

3.2.3 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 3.

3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le 125°C$	Device type	Limits Min	Limits Max	Unit
High level output voltage	V_{OH}	V_{CC} = 4.5 V; I_{OH} = -1 mA; V_{IL} = .8 V; V_{IH} = 2.0 V	All	2.5		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V; I_{OL} = 20 mA; V_{IL} = .8 V; V_{IH} = 2.0 V	All		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; I_{IN} = -18 mA; T_C = 25°C	All		-1.2	V
High level input current	I_{IH1}	V_{CC} = 5.5 V; V_{IH} = 2.7 V	All		20	µA
	I_{IH2}	V_{CC} = 5.5 V; V_{IH} = 7.0 V	All		100	µA
Low level input current	I_{IL1}	V_{CC} = 5.5 V; V_{IL} = .5 V	All	-.03	-.60	mA
Short circuit output current 1/	I_{OS}	V_{CC} = 5.5 V; V_{OS} = 0 V	All	-60	-150	mA
Output drive	I_{OD}	V_{CC} = 4.5 V; V_{IN} = 5.5 V	All	60		mA
Supply current	I_{CC}	V_{CC} = 5.5 V	01		20	mA
			02		20	mA
Propagation delay time low to high level		V_{CC} = 5.0 V, (see figure 5) C_L = 50 pF ±10%				
$\overline{E}$ to $\overline{O}$	t_{PLH1}		01	3.5	11.0	ns
			02	3.0	9.0	ns
E to $\overline{O}$	t_{PLH2}		01	4.0	12.5	ns
A to $\overline{O}$	t_{PLH3}		01	3.5	12.0	ns
			02	2.5	12.5	ns
Propagation delay time high to low level		V_{CC} = 5.0 V, (see figure 5) C_L = 50 pF ±10%				
$\overline{E}$ to $\overline{O}$	t_{PHL1}		01	3.0	8.0	ns
			02	2.5	8.0	ns
E to $\overline{O}$	t_{PHL2}		01	3.5	8.5	ns
A to $\overline{O}$	t_{PHL3}		01	4.0	9.5	ns
			02	3.5	9.5	ns

1/ Not more than one output should be shorted at a time.

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3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

TABLE II. Electrical tests requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7,8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for Group C periodic inspections	N/A	None
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

* PDA applies to subgroup 1 (see 4.2c).

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container:

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 11 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883 and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

1. Test condition D or E, using the circuit shown on figure 4, or equivalent.

2. T_A = 125°C minimum.

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b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The PDA for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady state life test (method 1005 of MIL-STD-883) conditions:

1. Test condition D or E, using the circuit shown on figure 4 or equivalent.

2. T_A = 125°C, minimum.

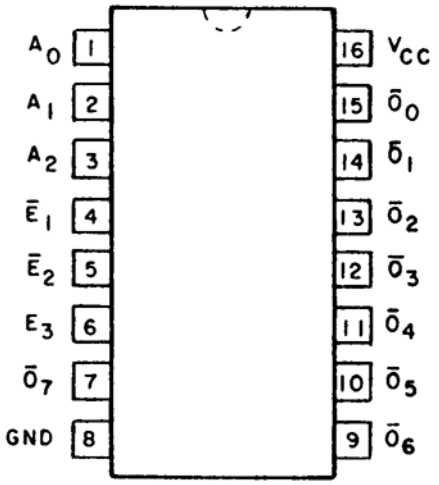
3. Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table III.

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Device type 01

Case E and F



Device type 02

Case E and F

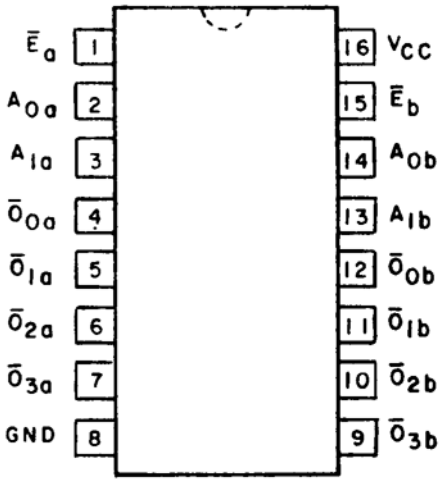
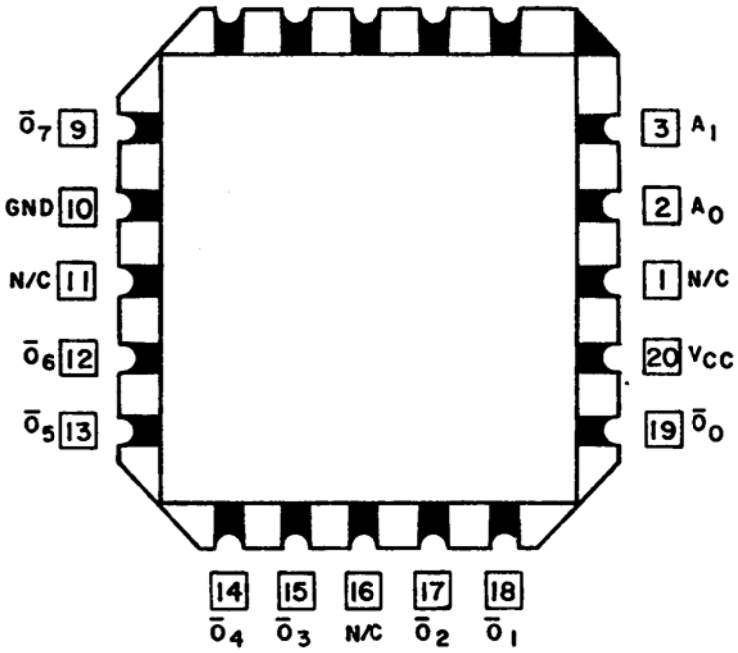


FIGURE 1. Terminal connections.

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Device 01
Case X



Device 02
Case X

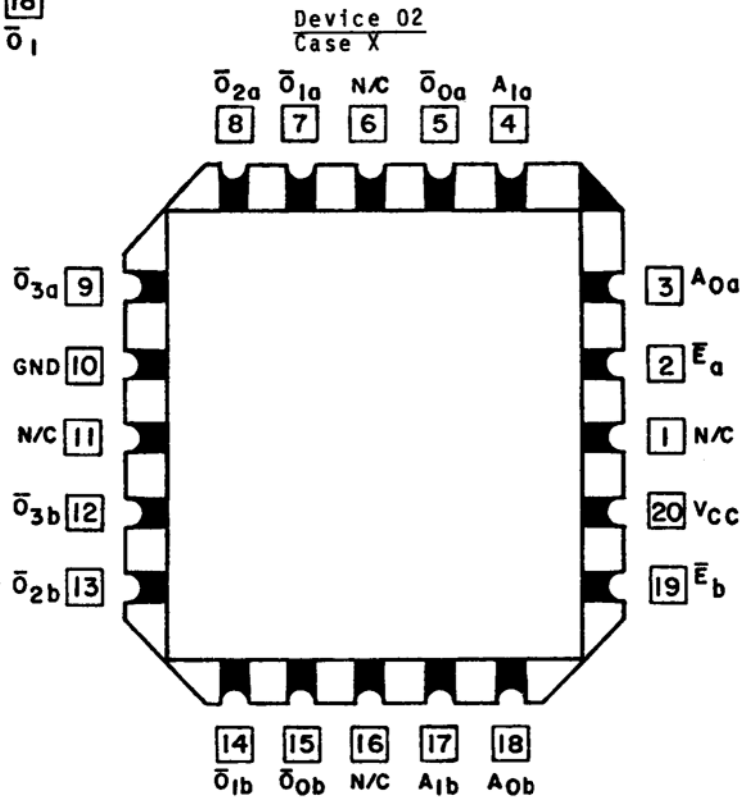
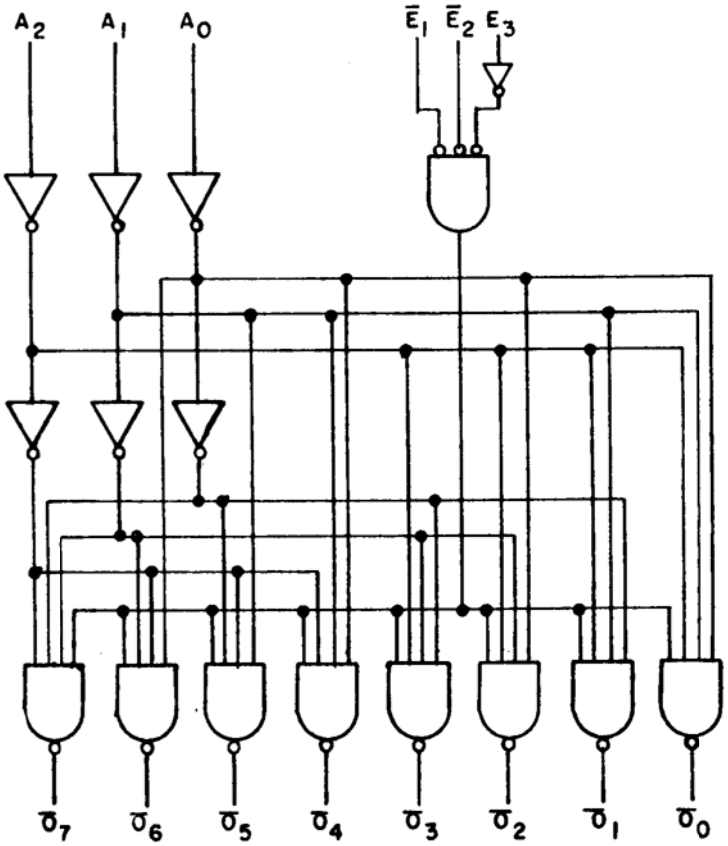


FIGURE 1. Terminal Connections - Continued.

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Device type 01



Device type 02

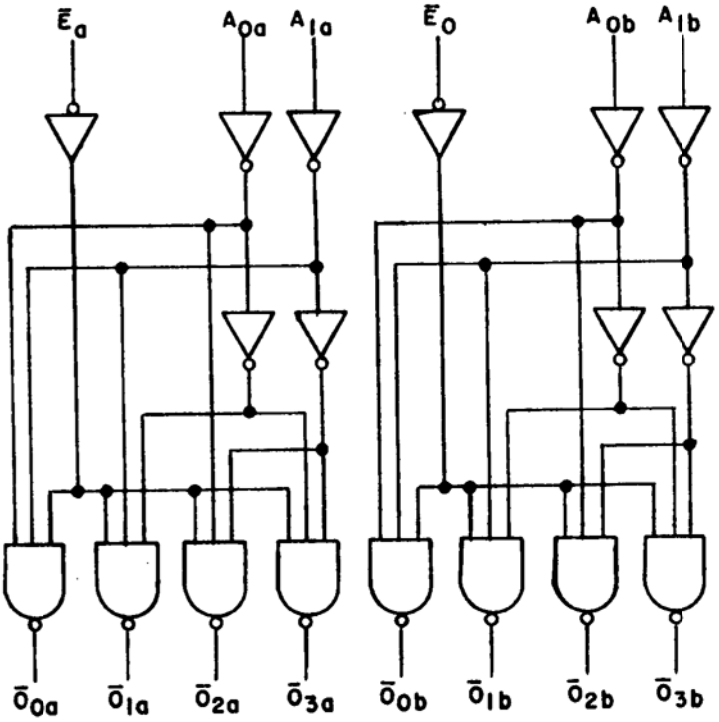


FIGURE 2. Logic diagrams.

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Device type 01

INPUTS						OUTPUTS							
$\overline{E}_1$	$\overline{E}_2$	E_3	A_0	A_1	A_2	$\overline{O}_0$	$\overline{O}_1$	$\overline{O}_2$	$\overline{O}_3$	$\overline{O}_4$	$\overline{O}_5$	$\overline{O}_6$	$\overline{O}_7$
H	X	X	X	X	X	H	H	H	H	H	H	H	H
X	H	X	X	X	X	H	H	H	H	H	H	H	H
X	X	L	X	X	X	H	H	H	H	H	H	H	H
L	L	H	L	L	L	L	H	H	H	H	H	H	H
L	L	H	H	L	L	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	L	H	H	H	H	H
L	L	H	H	H	L	H	H	H	L	H	H	H	H
L	L	H	L	L	H	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	L	H	H
L	L	H	L	H	H	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	L

H = HIGH Voltage Level
L = LOW Voltage Level
X = Immaterial

Device type 02

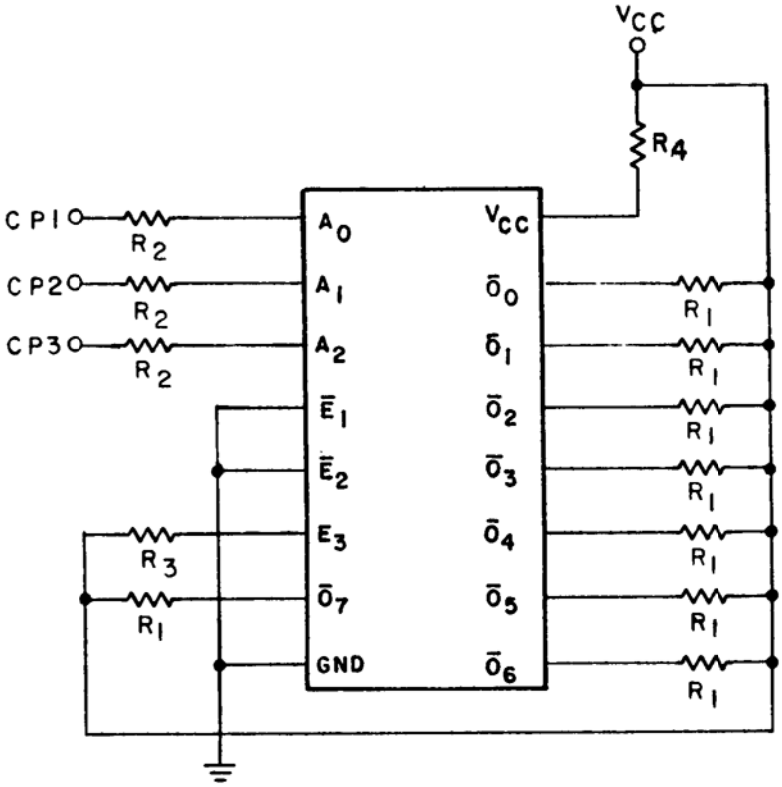
INPUTS			OUTPUTS			
$\overline{E}$	A_0	A_1	$\overline{O}_0$	$\overline{O}_1$	$\overline{O}_2$	$\overline{O}_3$
H	X	X	H	H	H	H
L	L	L	L	H	H	H
L	H	L	H	L	H	H
L	L	H	H	H	L	H
L	H	H	H	H	H	L

H = HIGH Voltage Level
L = LOW Voltage Level
X = Immaterial

FIGURE 3. Truth tables.

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Device type 01

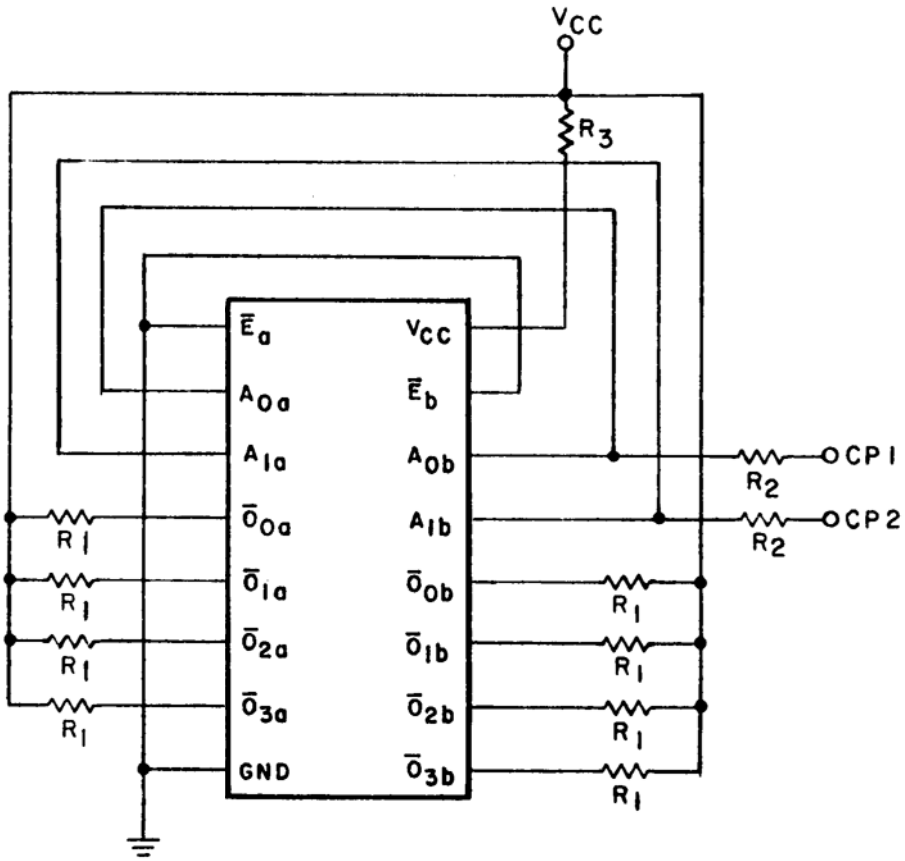


NOTES:
1. CP1 = 100 kHz ±50% square wave; duty cycle = 50 ±15%; CP2 = CP1 ÷ 2, CP3 = CP2 ÷ 2. V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω maximum; R2 = 51 kΩ ±5%; R3 = 1 kΩ maximum.
3. V_{CC} and R_4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit.

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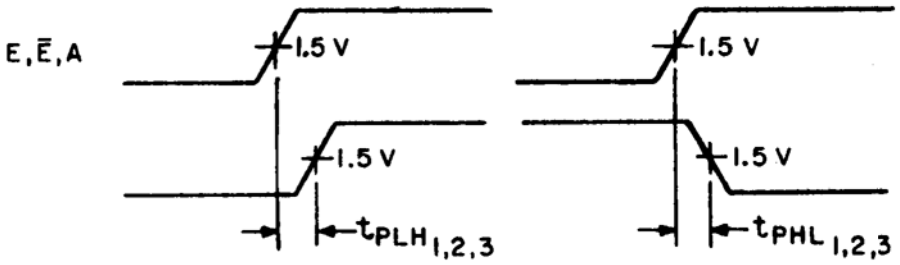
Device type 02



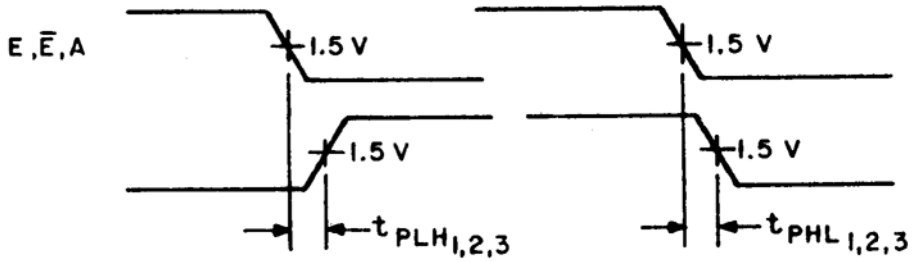
NOTES:
1. CP1 = 100 kHz ±50% square wave; duty cycle = 50 ±15%; CP2 = CP1 ÷ 2. V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω maximum; R2 = 51 kΩ ±5%.
3. V_{CC} and R_3 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit - Continued.

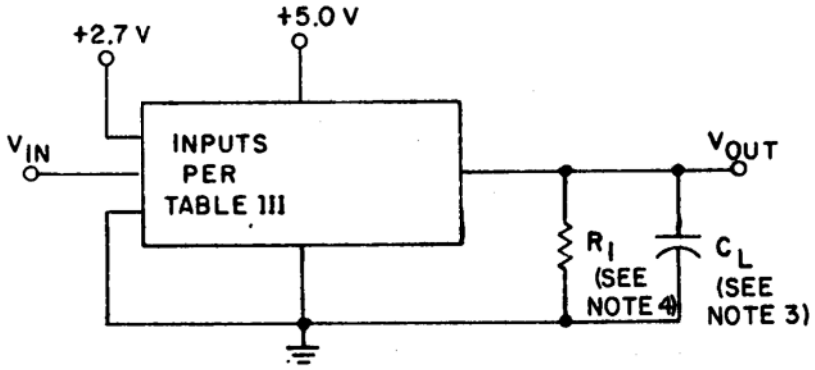
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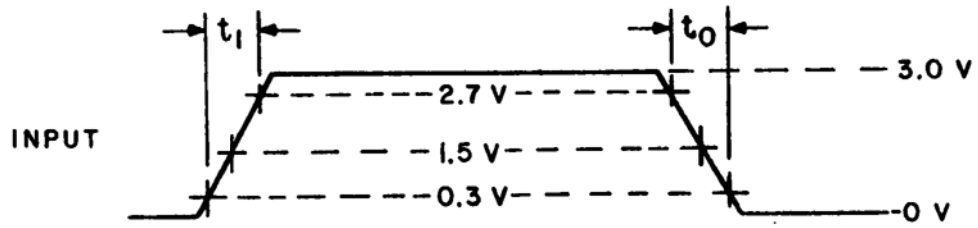
Device types 01 and 02



Device types 01 and 02



TEST CIRCUIT



NOTES:
1. V_{IN} = Input pulse and has the following characteristics:

   $t_1 = t_0 \leq 2.5$ ns, PRR $\leq$ 1 MHz, $Z_{OUT} \approx 50\Omega$.

2. Inputs not under test are at ground.
3. C_L = 50 pF ±10%, including scope probe, wiring and stray capacitance without package in test fixture.
4. R_1 = 499Ω ±5%.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Switching time waveforms (circuit for all device types).

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤.8 V, or open)

Subgroup	Symbol	MIL-STD-883-method	Case X 1/ E, F	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	A_0	A_1	A_2	$\overline{E}_1$	$\overline{E}_2$	E_3	$\overline{O}_7$	GND	$\overline{O}_6$	$\overline{O}_5$	$\overline{O}_4$	$\overline{O}_3$	$\overline{O}_2$	$\overline{O}_1$	$\overline{O}_0$	V_{CC}				
1 T_C = 25°C	V_{OL}	3007	1	2.0 V	2.0 V	2.0 V	.8 V	.8 V	2.0 V	20 mA	GND	20 mA							4.5 V	$\overline{O}_7$		0.5	V
	"	"	2	.8 V	2.0 V	"	"	"	"	"	"		20 mA						"	$\overline{O}_6$		"	"
	"	"	3	2.0 V	.8 V	"	"	"	"	"	"			20 mA					"	$\overline{O}_5$		"	"
	"	"	4	.8 V	.8 V	"	"	"	"	"	"				20 mA				"	$\overline{O}_4$		"	"
	"	"	5	2.0 V	2.0 V	.8 V	"	"	"	"	"					20 mA			"	$\overline{O}_3$		"	"
	"	"	6	.8 V	2.0 V	"	"	"	"	"	"						20 mA		"	$\overline{O}_2$		"	"
	"	"	7	2.0 V	.8 V	"	"	"	"	"	"							20 mA	"	$\overline{O}_1$		"	"
	"	"	8	.8 V	.8 V	"	"	"	"	"	"								"	$\overline{O}_0$		"	"
	V_{OH}	3006	9	5.5 V	5.5 V	5.5 V	0 V	0 V	.8 V	-1 mA	"								"	$\overline{O}_7$	2.5		"
	"	"	10	"	"	"	0 V	2.0 V	5.5 V	"	"								"	"	"		"
	"	"	11	"	"	"	2.0 V	0 V	"	"	"								"	$\overline{O}_6$	"		"
	"	"	12	2.0 V	.8 V	2.0 V	0 V	"	"	"	"	-1 mA							"	$\overline{O}_5$	"		"
	"	"	13	"	2.0 V	"	"	"	"	"	"		-1 mA						"	$\overline{O}_4$	"		"
	"	"	14	"	2.0 V	"	"	"	"	"	"			-1 mA					"	$\overline{O}_3$	"		"
	"	"	15	.8 V	.8 V	.8 V	"	"	"	"	"				-1 mA				"	$\overline{O}_2$	"		"
	"	"	16	2.0 V	2.0 V	2.0 V	"	"	"	"	"					-1 mA			"	$\overline{O}_1$	"		"
	"	"	17	.8 V	.8 V	.8 V	"	"	"	"	"						-1 mA		"	$\overline{O}_0$	"		"
	"	"	18	"	.8 V	"	"	"	"	"	"							-1 mA	"	"	"		"
	"	"	19	"	2.0 V	"	"	"	"	"	"								"	"	"		"
	"	"	20	0 V	0 V	0 V	"	"	.8 V	"	"								"	"	"		"
	"	"	21	"	"	"	"	2.0 V	5.5 V	"	"								"	"	"		"
	"	"	22	"	"	"	2.0 V	0 V	5.5 V	"	"								"	"	"		"
	V_{IC}		23	-18 mA							"								"	A_0		-1.2	"
	"		24		-18 mA						"								"	A_1		"	"
	"		25			-18 mA					"								"	A_2		"	"
	"		26				-18 mA				"								"	$\overline{E}_1$		"	"
	"		27					-18 mA			"								"	$\overline{E}_2$		"	"
	"		28						-18 mA		"								"	E_3		"	"
	I_{IH1}	3010	29	2.7 V							"								5.5 V	A_0		20	µA
	"	"	30		2.7 V						"								"	A_1		"	"
	"	"	31			2.7 V					"								"	A_2		"	"
	"	"	32				2.7 V				"								"	$\overline{E}_1$		"	"
	"	"	33					2.7 V			"								"	$\overline{E}_2$		"	"
	"	"	34						2.7 V		"								"	E_3		"	"
	I_{IH2}	"	35	7.0 V							"								"	A_0		100	"
	"	"	36		7.0 V						"								"	A_1		"	"
	"	"	37			7.0 V					"								"	A_2		"	"
	"	"	38				7.0 V				"								"	$\overline{E}_1$		"	"
	"	"	39					7.0 V			"								"	$\overline{E}_2$		"	"
	"	"	40						7.0 V		"								"	E_3		"	"
	I_{IL}	3009	41	.5 V							"								"	A_0	2/	2/	mA
	"	"	42		.5 V						"								"	A_1	"	"	"
	"	"	43			.5 V					"								"	A_2	"	"	"
	"	"	44				.5 V				"								"	$\overline{E}_1$	"	"	"
	"	"	45					.5 V			"								"	$\overline{E}_2$	"	"	"
	I_{IL1}	"	46						.5 V		"								"	E_3	"	"	"
	I_{OS}	3011	47	5.5 V	5.5 V	5.5 V	0 V	0 V	0 V	0 V	"								"	$\overline{O}_7$	-60	-150	"
	"	"	48	0 V	5.5 V	"	"	"	"	"	"	0 V							"	$\overline{O}_6$	"	"	"
	"	"	49	5.5 V	0 V	"	"	"	"	"	"		0 V						"	$\overline{O}_5$	"	"	"
	"	"	50	0 V	0 V	"	"	"	"	"	"			0 V					"	$\overline{O}_4$	"	"	"
	"	"	51	5.5 V	5.5 V	0 V	"	"	"	"	"				0 V				"	$\overline{O}_3$	"	"	"
	"	"	52	0 V	5.5 V	"	"	"	"	"	"					0 V			"	$\overline{O}_2$	"	"	"
	"	"	53	5.5 V	0 V	"	"	"	"	"	"						0 V		"	$\overline{O}_1$	"	"	"
	"	"	54	0 V	0 V	"	"	"	"	"	"							0 V	"	$\overline{O}_0$	"	"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤.8 V, or open)

Subgroup	Symbol	MIL-STD-883-method	Case X 1/ E, F	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
				A_0	A_1	A_2	$\overline{E}_1$	$\overline{E}_2$	E_3	$\overline{O}_7$	GND	$\overline{O}_6$	$\overline{O}_5$	$\overline{O}_4$	$\overline{O}_3$	$\overline{O}_2$	$\overline{O}_1$	$\overline{O}_0$	V_{CC}				
T_C = 25°C	I_{OD}		55	5.5 V	5.5 V	5.5 V	0.0 V	0.0 V	5.5 V	2.5 V	GND	2.5 V							4.5 V	$\overline{O}_7$	60		mA
	"		56	0.0 V	5.5 V	"	"	"	"		"		2.5 V						"	$\overline{O}_6$	"		"
	"		57	5.5 V	0.0 V	"	"	"	"		"			2.5 V					"	$\overline{O}_5$	"		"
	"		58	0.0 V	0.0 V	"	"	"	"		"				2.5 V				"	$\overline{O}_4$	"		"
	"		59	5.5 V	5.5 V	0.0 V	"	"	"		"					2.5 V			"	$\overline{O}_3$	"		"
	"		60	0.0 V	5.5 V	"	"	"	"		"						2.5 V		"	$\overline{O}_2$	"		"
	"		61	5.5 V	0.0 V	"	"	"	"		"							2.5 V	"	$\overline{O}_1$	"		"
	"		62	0.0 V	0.0 V	"	"	"	"		"								"	$\overline{O}_0$	"		"
	I_{CC}	3005	63	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"		"								5.5 V	V_{CC}		20	"
2 T_C = +125°C	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3 T_C = -55°C	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
7 T_C = 25°C	Functional tests 3/	3014	64	B	B	B	B	B	A	H	GND	H	H	H	H	H	H	L	L	4/	All outputs			
	"	"	65	A	B	"	"	"	"	"	"	"	"	"	"	H	L	H	"	"				
	"	"	66	B	A	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"				
	"	"	67	A	A	"	"	"	"	"	"	"	"	"	L	H	"	"	"	"				
	"	"	68	B	B	A	"	"	"	"	"	"	"	L	H	"	"	"	"	"				
	"	"	69	A	B	"	"	"	"	"	"	"	L	H	"	"	"	"	"	"				
	"	"	70	B	A	"	"	"	"	"	"	L	H	"	"	"	"	"	"	"				
	"	"	71	A	A	"	"	"	"	L	"	H	"	"	"	"	"	"	"	"				
	"	"	72	B	B	B	"	"	B	H	"	"	"	"	"	"	"	"	"	"				
	"	"	73	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
	"	"	74	"	A	B	"	A	"	"	"	"	"	"	"	"	"	"	"	"				
	"	"	75	"	A	A	"	A	A	"	"	"	"	"	"	"	"	"	"	"				
	"	"	76	A	B	B	A	B	B	"	"	"	"	"	"	"	"	"	"	"				
	"	"	77	"	B	A	"	B	A	"	"	"	"	"	"	"	"	"	"	"				
	"	"	78	"	A	B	"	A	B	"	"	"	"	"	"	"	"	"	"	"				
	"	"	79	"	A	A	"	A	A	"	"	"	"	"	"	"	"	"	"	"				
8	Repeat subgroup 7 tests at T_C = +125°C and T_C = -55°C.																						
9 T_C = 25°C	t_{PLH1}	3003 Figure 5	80	0 V	0 V	0 V	IN	0 V	2.7 V		GND							OUT	5.0 V	E_1 to $\overline{O}_0$	3.0	7.0	ns
	"	"	81	2.7 V	0 V	"	"	"	"		"							"	"	E_1 to $\overline{O}_1$	"	"	"
	"	"	82	0 V	2.7 V	"	"	"	"		"					OUT		"	"	E_1 to $\overline{O}_2$	"	"	"
	"	"	83	2.7 V	2.7 V	"	"	"	"		"				OUT			"	"	E_1 to $\overline{O}_3$	"	"	"
	"	"	84	0 V	0 V	2.7 V	"	"	"		"			OUT				"	"	E_1 to $\overline{O}_4$	"	"	"
	"	"	85	2.7 V	0 V	"	"	"	"		"		OUT					"	"	E_1 to $\overline{O}_5$	"	"	"
	"	"	86	0 V	2.7 V	"	"	"	"		"	OUT						"	"	E_1 to $\overline{O}_6$	"	"	"
	"	"	87	2.7 V	2.7 V	"	"	"	"	OUT	"							"	"	E_1 to $\overline{O}_7$	"	"	"
	"	"	88	0 V	0 V	0 V	0 V	IN	"		"							OUT	"	E_2 to $\overline{O}_0$	"	"	"
	"	"	89	2.7 V	0 V	"	"	"	"		"						OUT		"	E_2 to $\overline{O}_1$	"	"	"
	"	"	90	0 V	2.7 V	"	"	"	"		"					OUT			"	E_2 to $\overline{O}_2$	"	"	"
	"	"	91	2.7 V	2.7 V	"	"	"	"		"				OUT				"	E_2 to $\overline{O}_3$	"	"	"
	"	"	92	0 V	0 V	2.7 V	"	"	"		"			OUT					"	E_2 to $\overline{O}_4$	"	"	"
	"	"	93	2.7 V	0 V	"	"	"	"		"		OUT						"	E_2 to $\overline{O}_5$	"	"	"
	"	"	94	0 V	2.7 V	"	"	"	"		"	OUT							"	E_2 to $\overline{O}_6$	"	"	"
	"	"	95	2.7 V	2.7 V	"	"	"	"	OUT	"								"	E_2 to $\overline{O}_7$	"	"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case X 1/ E, F	Test no.	1 A_0	2	3 A_1	4 A_2	5 E_1	7 E_2	8 E_3	9 O_7	10 GND	12 O_6	13 O_5	14 O_4	15 O_3	17 O_2	18 O_1	19 O_0	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = 25°C	t_{PHL1}	3003 Figure 5		96	0 V		0 V	0 V	IN	0 V	2.7		GND							OUT	5.0 V	I_{E1} to O_0	3.0	7.0	ns
				97	2.7 V		0 V	"	"	"	"		"								"	I_{E1} to O_1	"	"	"
				98	0 V		2.7 V	"	"	"	"		"								"	I_{E1} to O_2	"	"	"
				99	2.7 V		2.7 V	"	"	"	"		"								"	I_{E1} to O_3	"	"	"
				100	0 V		0 V	2.7 V	"	"	"		"			OUT					"	I_{E1} to O_4	"	"	"
				101	2.7 V		0 V	"	"	"	"		"			"					"	I_{E1} to O_5	"	"	"
				102	0 V		2.7 V	"	"	"	"		"			"					"	I_{E1} to O_6	"	"	"
				103	2.7 V		2.7 V	"	"	"	"	OUT	"	OUT		"					"	I_{E1} to O_7	"	"	"
				104	0 V		0 V	0 V	0 V	IN	"		"								"	I_{E2} to O_0	"	"	"
				105	2.7 V		0 V	"	"	"	"		"								"	I_{E2} to O_1	"	"	"
				106	0 V		2.7 V	"	"	"	"		"								"	I_{E2} to O_2	"	"	"
				107	2.7 V		2.7 V	"	"	"	"		"								"	I_{E2} to O_3	"	"	"
				108	0 V		0 V	2.7 V	"	"	"		"			OUT					"	I_{E2} to O_4	"	"	"
				109	2.7 V		0 V	"	"	"	"		"			"					"	I_{E2} to O_5	"	"	"
				110	0 V		2.7 V	"	"	"	"		"			"					"	I_{E2} to O_6	"	"	"
				111	2.7 V		2.7 V	"	"	"	"	OUT	"	OUT		"					"	I_{E2} to O_7	"	"	"
	t_{PLH2}	3003 Figure 5		112	0 V		0 V	0 V	"	0 V	IN		"							OUT	"	I_{E3} to O_0	4.0	8.0	"
				113	2.7 V		0 V	"	"	"	"		"								"	I_{E3} to O_1	"	"	"
				114	0 V		2.7 V	"	"	"	"		"								"	I_{E3} to O_2	"	"	"
				115	2.7 V		2.7 V	"	"	"	"		"								"	I_{E3} to O_3	"	"	"
				116	0 V		0 V	2.7 V	"	"	"		"			OUT					"	I_{E3} to O_4	"	"	"
				117	2.7 V		0 V	"	"	"	"		"			"					"	I_{E3} to O_5	"	"	"
				118	0 V		2.7 V	"	"	"	"		"			"					"	I_{E3} to O_6	"	"	"
				119	2.7 V		2.7 V	"	"	"	"	OUT	"	OUT		"					"	I_{E3} to O_7	"	"	"
	t_{PHL2}	"		120	0 V		0 V	0 V	"	"	"		"							OUT	"	I_{E3} to O_0	3.5	7.5	"
				121	2.7 V		0 V	"	"	"	"		"								"	I_{E3} to O_1	"	"	"
				122	0 V		2.7 V	"	"	"	"		"								"	I_{E3} to O_2	"	"	"
				123	2.7 V		2.7 V	2.7 V	"	"	"		"			OUT					"	I_{E3} to O_3	"	"	"
				124	0 V		0 V	"	"	"	"		"			"					"	I_{E3} to O_4	"	"	"
				125	2.7 V		0 V	"	"	"	"		"			"					"	I_{E3} to O_5	"	"	"
				126	0 V		2.7 V	"	"	"	"		"			"					"	I_{E3} to O_6	"	"	"
				127	2.7 V		2.7 V	"	"	"	"	OUT	"	OUT		"					"	I_{E3} to O_7	"	"	"
	t_{PLH3}	"		128	IN		0 V	0 V	"	"	2.7 V		"							OUT	"	A_0 to O_0	3.5	7.0	"
				129	"		0 V	"	"	"	"		"								"	A_0 to O_1	"	"	"
				130	"		2.7 V	"	"	"	"		"								"	A_0 to O_2	"	"	"
				131	"		2.7 V	"	"	"	"		"								"	A_0 to O_3	"	"	"
				132	"		0 V	2.7 V	"	"	"		"								"	A_0 to O_4	"	"	"
				133	"		0 V	"	"	"	"		"								"	A_0 to O_5	"	"	"
				134	"		2.7 V	"	"	"	"		"								"	A_0 to O_6	"	"	"
				135	"		2.7 V	"	"	"	"	OUT	"	OUT		"					"	A_0 to O_7	"	"	"
				136	0 V		IN	0 V	0 V	"	"		"								"	A_1 to O_0	"	"	"
				137	2.7 V		"	"	"	"	"		"								"	A_1 to O_1	"	"	"
				138	0 V		"	"	"	"	"		"								"	A_1 to O_2	"	"	"
				139	2.7 V		"	2.7 V	"	"	"		"								"	A_1 to O_3	"	"	"
				140	0 V		"	"	"	"	"		"								"	A_1 to O_4	"	"	"
				141	2.7 V		"	"	"	"	"		"								"	A_1 to O_5	"	"	"
				142	0 V		"	"	"	"	"		"								"	A_1 to O_6	"	"	"
				143	2.7 V		"	"	"	"	"	OUT	"	OUT		"					"	A_1 to O_7	"	"	"
				144	0 V		0 V	IN	"	"	"		"								"	A_2 to O_0	"	"	"
				145	2.7 V		0 V	"	"	"	"		"								"	A_2 to O_1	"	"	"
				146	0 V		2.7 V	"	"	"	"		"								"	A_2 to O_2	"	"	"
				147	2.7 V		2.7 V	"	"	"	"		"								"	A_2 to O_3	"	"	"
				148	0 V		0 V	"	"	"	"		"								"	A_2 to O_4	"	"	"
				149	2.7 V		0 V	"	"	"	"		"								"	A_2 to O_5	"	"	"
				150	0 V		2.7 V	"	"	"	"		"								"	A_2 to O_6	"	"	"
				151	2.7 V		2.7 V	"	"	"	"	OUT	"	OUT		"					"	A_2 to O_7	"	"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤.8 V, or open)

Subgroup	Symbol	MIL-STD-883-method	Case X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
			E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	A_0	A_1	A_2	$\overline{E}_1$	$\overline{E}_2$	E_3	$\overline{O}_7$	GND	$\overline{O}_6$	$\overline{O}_5$	$\overline{O}_4$	$\overline{O}_3$	$\overline{O}_2$	$\overline{O}_1$	$\overline{O}_0$	V_{CC}				
9 T_C = 25°C	t_{PHL3}	3003 Figure 5	152	IN	0 V	0 V	0 V	0 V	2.7 V		GND							OUT	5.0 V	A_0 to $\overline{O}_5$	4.0	8.0	ns
	"	"	153	"	0 V	"	"	"	"		"					OUT		"	"	A_0 to $\overline{O}_1$	"	"	"
	"	"	154	"	2.7 V	"	"	"	"		"						OUT	"	"	A_0 to $\overline{O}_3$	"	"	"
	"	"	155	"	2.7 V	"	"	"	"		"				OUT			"	"	A_0 to $\overline{O}_2$	"	"	"
	"	"	156	"	0 V	2.7 V	"	"	"		"			OUT				"	"	A_0 to $\overline{O}_4$	"	"	"
	"	"	157	"	0 V	"	"	"	"		"		OUT					"	"	A_0 to $\overline{O}_6$	"	"	"
	"	"	158	"	2.7 V	"	"	"	"		"	OUT						"	"	A_0 to $\overline{O}_7$	"	"	"
	"	"	159	"	2.7 V	"	"	"	"	OUT	"							"	"	A_0 to GND	"	"	"
	"	"	160	0 V	IN	0 V	"	"	"		"							OUT	"	A_1 to $\overline{O}_5$	"	"	"
	"	"	161	2.7 V	"	"	"	"	"		"					OUT		"	"	A_1 to $\overline{O}_1$	"	"	"
	"	"	162	0 V	"	"	"	"	"		"						OUT	"	"	A_1 to $\overline{O}_3$	"	"	"
	"	"	163	2.7 V	"	"	"	"	"		"				OUT			"	"	A_1 to $\overline{O}_2$	"	"	"
	"	"	164	0 V	"	2.7 V	"	"	"		"			OUT				"	"	A_1 to $\overline{O}_4$	"	"	"
	"	"	165	2.7 V	"	"	"	"	"		"		OUT					"	"	A_1 to $\overline{O}_6$	"	"	"
	"	"	166	0 V	"	"	"	"	"		"	OUT						"	"	A_1 to $\overline{O}_7$	"	"	"
	"	"	167	2.7 V	"	"	"	"	"	OUT	"							"	"	A_1 to GND	"	"	"
	"	"	168	0 V	0 V	IN	"	"	"		"							OUT	"	A_2 to $\overline{O}_5$	"	"	"
	"	"	169	2.7 V	0 V	"	"	"	"		"						OUT	"	"	A_2 to $\overline{O}_1$	"	"	"
	"	"	170	0 V	2.7 V	"	"	"	"		"					OUT		"	"	A_2 to $\overline{O}_3$	"	"	"
	"	"	171	2.7 V	2.7 V	"	"	"	"		"				OUT			"	"	A_2 to $\overline{O}_2$	"	"	"
	"	"	172	0 V	0 V	"	"	"	"		"			OUT				"	"	A_2 to $\overline{O}_4$	"	"	"
	"	"	173	2.7 V	0 V	"	"	"	"		"		OUT					"	"	A_2 to $\overline{O}_6$	"	"	"
	"	"	174	0 V	2.7 V	"	"	"	"		"	OUT						"	"	A_2 to $\overline{O}_7$	"	"	"
	"	"	175	2.7 V	2.7 V	"	"	"	"	OUT	"							"	"	A_2 to GND	"	"	"
10 T_C = +125°C	t_{PLH1} t_{PHL1} t_{PLH2} t_{PHL2} t_{PLH3} t_{PHL3}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits as shown.																			3.5 3.0 4.0 3.5 3.5 4.0	11.0 8.0 12.5 8.5 12.0 9.5	"
11 T_C = -55°C		Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																					

1/ For case X pins not referenced are N/C.

2/ I_{IL} limits (mA) min/max values for circuit shown:

Parameter	Test no.	Circuits A	Circuits B	Circuits C
I_{IL}	41 through 45	-.27/-.60	-.03/-.60	-.03/-.60
I_{IL1}	46	-.22/-.60	-.03/-.60	-.03/-.60

3/ A = 4.5 V, B = 0.0 V,
H ≥2.5 V, L ≤.5 V with high speed checker double comparator, or
H ≥1.5 V, L ≤1.5 V with high speed checker single comparator.

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

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TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case X E, F Test no.	1 E_a	2 A_{0a}	3 A_{1a}	4 σ_{0a}	5 σ_{1a}	6 σ_{2a}	7 σ_{3a}	8 GND	9 σ_{3b}	10 σ_{2b}	11 σ_{1b}	12 σ_{0b}	13 A_{1b}	14 A_{0b}	15 E_b	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OL}	3007	1	.8 V	.8 V	.8 V	20 mA	20 mA			GND								4.5 V	σ_{0a}		.5	V
	"	"	2	"	2.0 V	"					"								"	σ_{1a}		"	"
	"	"	3	"	.8 V	2.0 V					"								"	σ_{2a}		"	"
	"	"	4	"	2.0 V	2.0 V				20 mA	"	20 mA	20 mA			2.0 V	2.0 V	.8 V	"	σ_{3a}		"	"
	"	"	5	"							"			20 mA		.8 V	.8 V	"	"	σ_{3b}		"	"
	"	"	6	"							"					.8 V	2.0 V	"	"	σ_{2b}		"	"
	"	"	7	"							"					.8 V	.8 V	"	"	σ_{1b}		"	"
	"	"	8	"							"								"	σ_{0b}		"	"
	V_{OH}	3006	9	2.0 V	0 V	0 V	-1 mA				"								"	σ_{0a}	2.5		"
	"	"	10	0 V	2.0 V	0 V	-1 mA				"								"	σ_{1a}	"		"
	"	"	11	"	5.5 V	2.0 V		-1 mA			"								"	σ_{2a}	"		"
	"	"	12	"	0 V	.8 V		-1 mA			"								"	σ_{3a}	"		"
	"	"	13	"	.8 V	5.5 V					"	-1 mA				5.5 V	5.5 V	2.0 V	"	σ_{3b}	"		"
	"	"	14	2.0 V	5.5 V	5.5 V					"	-1 mA				5.5 V	.8 V	0 V	"	σ_{2b}	"		"
	"	"	15								"		-1 mA			.8 V	0 V	"	"	σ_{1b}	"		"
	"	"	16								"					2.0 V	5.5 V	"	"	σ_{0b}	"		"
	"	"	17								"			-1 mA	-1 mA	0 V	2.0 V	2.0 V	"	σ_{0b}	"		"
	"	"	18								"				-1 mA	0 V	0 V	"	"	σ_{1b}	"		"
	"	"	19								"								"	σ_{2b}	"		"
	"	"	20								"								"	σ_{3b}	"		"
	V_{IC}		21	-18 mA							"								"	E_a	-1.2		"
	"		22		-18 mA						"								"	A_{0a}	"		"
	"		23			-18 mA					"								"	A_{1a}	"		"
	"		24								"					-18 mA			"	A_{1b}	"		"
	"		25								"						-18 mA		"	A_{0b}	"		"
	"		26								"							-18 mA	"	E_b	"		"
	I_{IH1}	3010	27	2.7 V							"								5.5 V	E_a	20		µA
	"	"	28		2.7 V						"								"	A_{0a}	"		"
	"	"	29			2.7 V					"					2.7 V			"	A_{1a}	"		"
	"	"	30								"						2.7 V		"	A_{1b}	"		"
	"	"	31								"							2.7 V	"	A_{0b}	"		"
	"	"	32								"								"	E_b	"		"
	I_{IH2}	"	33	7.0 V							"								"	E_a	100		"
	"	"	34		7.0 V						"								"	A_{0a}	"		"
	"	"	35			7.0 V					"					7.0 V			"	A_{1a}	"		"
	"	"	36								"						7.0 V		"	A_{1b}	"		"
	"	"	37								"							7.0 V	"	A_{0b}	"		"
	"	"	38								"								"	E_b	"		"
	I_{IL1}	3009	39	.5 V							"								"	E_a	2/	2/	mA
	"	"	40		.5 V						"								"	A_{0a}	"	"	"
	"	"	41			.5 V					"					.5 V			"	A_{1a}	"	"	"
	"	"	42								"						.5 V		"	A_{1b}	"	"	"
	"	"	43								"							.5 V	"	A_{0b}	"	"	"
	"	"	44								"								"	E_b	"	"	"
	I_{OS}	3011	45	5.5 V	0 V	0 V	0 V				"								"	σ_{0a}	-60	-150	"
	"	"	46	"	"	"		0 V			"								"	σ_{1a}	"	"	"
	"	"	47	"	"	"			0 V		"								"	σ_{2a}	"	"	"
	"	"	48	"	"	"				0 V	"								"	σ_{3a}	"	"	"
	"	"	49								"	0 V							"	σ_{3b}	"	"	"
	"	"	50								"		0 V						"	σ_{2b}	"	"	"
	"	"	51								"			0 V					"	σ_{1b}	"	"	"
	"	"	52								"				0 V				"	σ_{0b}	"	"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤.8 V, or open)

Subgroup	Symbol	MIL-STD-883-method	Case X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
			E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	$\overline{E}_a$	A_{0a}	A_{1a}	$\overline{O}_{0a}$	$\overline{O}_{1a}$	$\overline{O}_{2a}$	$\overline{O}_{3a}$	GND	$\overline{O}_{3b}$	$\overline{O}_{2b}$	$\overline{O}_{1b}$	$\overline{O}_{0b}$	A_{1b}	A_{0b}	$\overline{E}_b$	V_{CC}				
T_C = 25°C	I_{OD}		53	0.0 V	0.0 V	0.0 V	2.5 V				GND								4.5 V	$\overline{O}_{0a}$	60		mA
			54	"	5.5 V	0.0 V		2.5 V			"								"	$\overline{O}_{1a}$	"		"
			55	"	0.0 V	5.5 V			2.5 V		"								"	$\overline{O}_{2a}$	"		"
			56	"	5.5 V	5.5 V				2.5 V	"								"	$\overline{O}_{3a}$	"		"
			57								"	2.5 V				5.5 V	5.5 V	0.0 V	"	$\overline{O}_{3b}$	"		"
			58								"		2.5 V			5.5 V	0.0 V	0.0 V	"	$\overline{O}_{2b}$	"		"
			59								"			2.5 V		0.0 V	5.5 V	0.0 V	"	$\overline{O}_{1b}$	"		"
			60								"				2.5 V	0.0 V	0.0 V	0.0 V	"	$\overline{O}_{0b}$	"		"
	I_{CC}		61	0 V	5.5 V	5.5 V					"					5.5 V	5.5 V	0 V	5.5 V	V_{CC}		20	"
2 T_C = +125°C	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3 T_C = -55°C	Same tests, terminal conditions, and limits as for subgroup 2, except T_C = -55°C.																						
7 T_C = 25°C	Functional tests 3/	3014	62	B	B	B	L	H	H	H	GND	H	H	H	L	B	B	B	4/	All outputs			
	"	"	63	"	A	A	H	L	H	"	"	"	L	L	"	A	A	"	"	"			
	"	"	64	"	B	A	L	H	L	"	"	"	H	H	"	B	A	"	"	"			
	"	"	65	"	A	A	H	L	H	"	"	"	L	H	"	A	A	"	"	"			
	"	"	66	A	B	B	L	H	L	"	"	"	H	H	"	B	B	A	"	"			
	"	"	67	"	A	A	H	L	H	"	"	"	L	H	"	A	B	"	"	"			
	"	"	68	"	B	A	L	H	L	"	"	"	H	H	"	B	A	"	"	"			
	"	"	69	"	A	B	H	L	H	"	"	"	L	H	"	A	B	"	"	"			
8	Repeat subgroup 7 tests at T_C = +125°C and T_C = -55°C.																						
9 T_C = 25°C	$t_{PLH}1$	3003 Figure 5	70	IN	0 V	0 V	OUT				GND								5.0 V	E_a to $\overline{O}_{0a}$	3.5	7.0	ns
	"	"	71	"	2.7 V	0 V		OUT			"								"	E_a to $\overline{O}_{1a}$	"	"	"
	"	"	72	"	0 V	2.7 V			OUT		"								"	E_a to $\overline{O}_{2a}$	"	"	"
	"	"	73	"	2.7 V	2.7 V				OUT	"								"	E_a to $\overline{O}_{3a}$	"	"	"
	"	"	74								"				OUT				"	E_b to $\overline{O}_{0b}$	"	"	"
	"	"	75								"			OUT				IN	"	E_b to $\overline{O}_{1b}$	"	"	"
	"	"	76								"		OUT			0 V	0 V	"	"	E_b to $\overline{O}_{2b}$	"	"	"
	"	"	77								"	OUT				2.7 V	2.7 V	"	"	E_b to $\overline{O}_{3b}$	"	"	"
	$t_{PHL}1$	"	78	IN	0 V	0 V	OUT				"								"	E_a to $\overline{O}_{0a}$	3.0	6.5	"
	"	"	79	"	2.7 V	0 V		OUT			"								"	E_a to $\overline{O}_{1a}$	"	"	"
	"	"	80	"	0 V	2.7 V			OUT		"								"	E_a to $\overline{O}_{2a}$	"	"	"
	"	"	81	"	2.7 V	2.7 V				OUT	"								"	E_a to $\overline{O}_{3a}$	"	"	"
	"	"	82								"				OUT				"	E_b to $\overline{O}_{0b}$	"	"	"
	"	"	83								"			OUT				IN	"	E_b to $\overline{O}_{1b}$	"	"	"
	"	"	84								"		OUT			0 V	0 V	"	"	E_b to $\overline{O}_{2b}$	"	"	"
	"	"	85								"	OUT				2.7 V	2.7 V	"	"	E_b to $\overline{O}_{3b}$	"	"	"
	$t_{PLH}3$	"	86	0 V	IN	0 V	OUT				"								"	A_{0a} to $\overline{O}_{0a}$	3.5	7.0	"
	"	"	87	"	"	0 V		OUT			"								"	A_{0a} to $\overline{O}_{1a}$	"	"	"
	"	"	88	"	"	2.7 V			OUT		"								"	A_{0a} to $\overline{O}_{2a}$	"	"	"
	"	"	89	"	"	2.7 V				OUT	"								"	A_{0a} to $\overline{O}_{3a}$	"	"	"
	"	"	90								"				OUT				"	A_{0b} to $\overline{O}_{0b}$	"	"	"
	"	"	91								"			OUT				0 V	"	A_{0b} to $\overline{O}_{1b}$	"	"	"
	"	"	92								"		OUT			0 V	IN	"	"	A_{0b} to $\overline{O}_{2b}$	"	"	"
	"	"	93								"	OUT				2.7 V	"	"	"	A_{0b} to $\overline{O}_{3b}$	"	"	"
	"	"	94	0 V	0 V	IN	OUT				"								"	A_{1a} to $\overline{O}_{0a}$	"	"	"
	"	"	95	"	2.7 V	"		OUT			"								"	A_{1a} to $\overline{O}_{1a}$	"	"	"
	"	"	96	"	0 V	"			OUT		"								"	A_{1a} to $\overline{O}_{2a}$	"	"	"
	"	"	97	"	2.7 V	"				OUT	"								"	A_{1a} to $\overline{O}_{3a}$	"	"	"
	"	"	98								"				OUT				"	A_{1b} to $\overline{O}_{0b}$	"	"	"
	"	"	99								"			OUT				0 V	"	A_{1b} to $\overline{O}_{1b}$	"	"	"
	"	"	100								"		OUT			IN	0 V	"	"	A_{1b} to $\overline{O}_{2b}$	"	"	"
	"	"	101								"	OUT				"	"	"	"	A_{1B} to $\overline{O}_{3b}$	"	"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤.8 V, or open)

Subgroup	Symbol	MIL-STD-883-method	Case X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
			E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	$\overline{E}_a$	A_{0a}	A_{1a}	$\overline{O}_{0a}$	$\overline{O}_{1a}$	$\overline{O}_{2a}$	$\overline{O}_{3a}$	GND	$\overline{O}_{3b}$	$\overline{O}_{2b}$	$\overline{O}_{1b}$	$\overline{O}_{0b}$	A_{1b}	A_{0b}	$\overline{E}_b$	V_{CC}				
9 T_C = 25°C	t_{PHL3}	3003 Figure 5	102	0 V	IN	0 V	OUT				GND								5.0 V	A_{0a} to $\overline{O}_{0a}$	4.0	8.0	ns
	"	"	103	"	"	0 V		OUT			"								"	A_{0a} to $\overline{O}_{1a}$	"	"	"
	"	"	104	"	"	2.7 V			OUT		"								"	A_{0a} to $\overline{O}_{2a}$	"	"	"
	"	"	105	"	"	2.7 V				OUT	"				OUT	0 V	IN	0 V	"	A_{0a} to $\overline{O}_{3a}$	"	"	"
	"	"	106								"			OUT		0 V	"	"	"	A_{0b} to $\overline{O}_{0b}$	"	"	"
	"	"	107								"		OUT			2.7 V	"	"	"	A_{0b} to $\overline{O}_{1b}$	"	"	"
	"	"	108								"	OUT				2.7 V	"	"	"	A_{0b} to $\overline{O}_{2b}$	"	"	"
	"	"	109								"								"	A_{0b} to $\overline{O}_{3b}$	"	"	"
	"	"	110	0 V	0 V	IN	OUT				"								"	A_{1a} to $\overline{O}_{0a}$	"	"	"
	"	"	111	"	2.7 V	"		OUT			"								"	A_{1a} to $\overline{O}_{1a}$	"	"	"
	"	"	112	"	0 V	"			OUT		"								"	A_{1a} to $\overline{O}_{2a}$	"	"	"
	"	"	113	"	2.7 V	"				OUT	"				OUT	IN	0 V	0 V	"	A_{1a} to $\overline{O}_{3a}$	"	"	"
	"	"	114								"			OUT		"	2.7 V	"	"	A_{1b} to $\overline{O}_{0b}$	"	"	"
	"	"	115								"		OUT			"	0 V	"	"	A_{1b} to $\overline{O}_{1b}$	"	"	"
	"	"	116								"	OUT				"	2.7 V	"	"	A_{1b} to $\overline{O}_{2b}$	"	"	"
	"	"	117								"								"	A_{1b} to $\overline{O}_{3b}$	"	"	"
10 T_C = +125°C	t_{PLH1}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits as shown.																			3.0	9.0	"
	t_{PHL1}																				2.5	8.0	"
	t_{PLH3}																				2.5	12.5	"
	t_{PHL3}																				3.5	9.5	"
11 T_C = -55°C		Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																					

1/ For case X pins not referenced are N/C.

2/ I_{IL} limits (mA) min/max values for circuit shown:

Parameter	Test no.	Circuits A	Circuits B	Circuits C
I_{IL}	39 through 44	-.25/-.60	-.03/-.60	-.03/-.60

3/ A = 4.5 V, B = 0.0 V, H ≥2.5 V, L ≤.5 V with high speed checker double comparator, or H ≥1.5, L ≤1.5 V with high speed checker single comparator.

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuits ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirement for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential
V_{IN}	Voltage level at an input terminal
V_{IC}	Input clamp voltage
I_{IN}	Current flowing into an input terminal

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired to Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F138
02	54F139

6.7 Manufacturers' designations. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designation.

Device type	Circuits A Fairchild	B Motorola	C Signetics
01	X	X	
02	X	X	

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/337A	MICROCIRCUITS, DIGITAL, ADVANCED SCHOTTKY, TTL, DECODERS

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS (Street, City, State, ZIP Code)

4. TYPE OF ORGANIZATION (Mark one)

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER (Specify): _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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